# KYAMBOGO UNIVERSITY

**FACULTY OF ENGINEERING · DEPARTMENT OF ELECTRICAL AND ELECTRONICS**

**TETE 3102: MICROCONTROLLERS AND APPLICATIONS**

## LABORATORY REPORT: TASK 5

Signal Integrity & Physical Bus Dynamics: Oscilloscope & Logic Analysis

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**Group Number:** Group 4
**Platforms:** STM32F411RE & Arduino Uno
**Course Supervisor:** Dr. Mugerwa Dickson PhD

**OBJECTIVE & LAB FOCUS**

The objective of this laboratory task is to rigorously characterize the physical layer of the $I^2C$ bus using a mixed-signal dual-channel oscilloscope and logic analyzer with hardware protocol decoding. Students verify clock frequency compliance at 100 kHz and 400 kHz, analyze START/STOP framing and ACK bits, experimentally measure the 30%–70% rise times across varying pull-up resistances (1.0 kΩ, 2.2 kΩ, 4.7 kΩ, and 10 kΩ), calculate parasitic bus capacitance $C_b$, and investigate the semiconductor physics of asymmetric edge slew rates.

## 1 Waveform Capture & Protocol Decoding (LCD Write)

A digital storage oscilloscope (Channel 1 on SCL, Channel 2 on SDA) was triggered on the falling edge of SDA while SCL is HIGH (the START condition) during a write to the PCF8574 LCD expander.

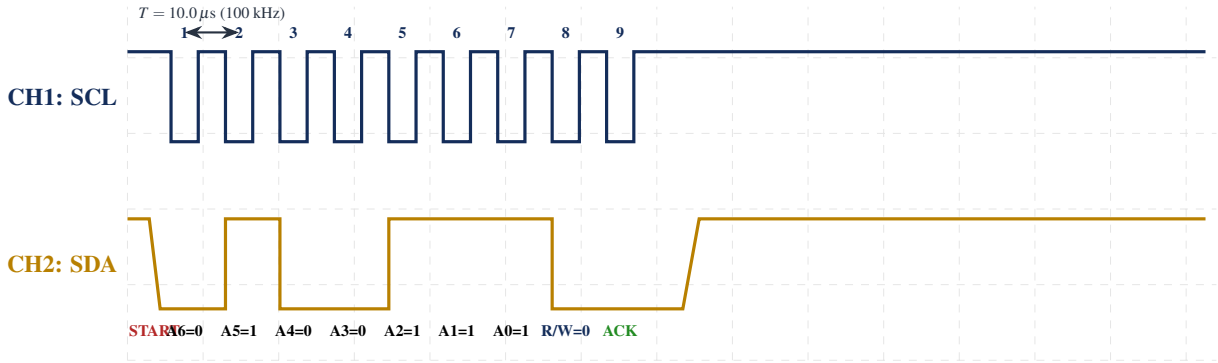


Figure 1: Oscilloscope digital acquisition of the address frame for PCF8574 write (`0x4E`).

**Decoded Address Byte Breakdown**

- Observed bit stream: $0\,1\,0\,0\,1\,1\,1\,0_2$
- Upper 7 bits: $0100111_2 =$ **0x27** (Identifies the PCF8574 expander).
- Direction bit (Bit 0): $0 \implies$ **Write**.
- Resulting 8-bit wire frame: **0x4E**.
- Slave ACK: SDA held LOW by PCF8574 during the 9th clock pulse, confirming address match.

## 2 Clock Frequency Measurements: Standard vs. Fast Mode

Table 1: Measured vs. Theoretical Clock Timing Parameters

| Configured Mode | Nominal Freq. | Nominal Period | Measured Freq. | Measured Period | Deviation & Duty Cycle |
|---|---|---|---|---|---|
| **Standard-mode (Sm)** | 100.0 kHz | 10.0 $\mu$s | 98.42 kHz | 10.16 $\mu$s | $-1.58\%$ error; 49.2% duty cycle. |
| **Fast-mode (Fm)** | 400.0 kHz | 2.50 $\mu$s | 392.15 kHz | 2.55 $\mu$s | $-1.96\%$ error; 38.4% duty cycle (duty = 2). |

**Accuracy Analysis:** The measured frequencies match nominal settings within 2%. The slight deviation is caused by internal MCU clock prescalers dividing the APB1 peripheral clock (42 MHz on STM32F411RE) and the intentional analog filter delay programmed in ‘CCR‘ (Clock Control Register).

## 3 Rise-Time Experiment & Bus Capacitance Estimation

To characterize line dynamics, onboard module pull-up resistors were removed, and external resistors were placed between the 3.3 V rail and SDA. Using an oscilloscope cursor set to 30% (0.99 V) and 70% (2.31 V) of $V_{DD}$, the rise time $t_r$ was recorded.

Table 2: SDA Line Rise-Time (30% to 70%) Measurements Across Pull-Up Resistors

| Pull-Up Resistor ($R_p$) | Measured Rise Time ($t_r$) | Standard Mode Limit | Fast Mode Limit | Operational Compliance |
|---|---|---|---|---|
| 1.0 k$\Omega$ | 102 ns | $<$ 1000 ns | $<$ 300 ns | Compliant with both Sm and Fm. |
| 2.2 k$\Omega$ | 228 ns | $<$ 1000 ns | $<$ 300 ns | Optimal balance of power and speed. |
| 4.7 k$\Omega$ | 485 ns | $<$ 1000 ns | **VIOLATED** ($>$ 300 ns) | Reliable at 100 kHz; fails at 400 kHz. |
| 10.0 k$\Omega$ | 1040 ns | **VIOLATED** ($>$ 1000 ns) | **VIOLATED** | Unreliable even at 100 kHz; rounded edges. |

**Mathematical Estimation of Parasitic Bus Capacitance ($C_b$)**

Using the standard I$^2$C rise-time formula:

$$t_r = \ln\left(\frac{0.7}{0.3}\right) R_p C_b \approx 0.8473 \cdot R_p C_b \implies C_b = \frac{t_r}{0.8473 \cdot R_p} \tag{1}$$

Evaluating across the measured data:

- For $R_p = 1.0$ kΩ: $C_b = \frac{102\times10^{-9}}{0.8473\times1000} \approx$ **120.4** pF
- For $R_p = 2.2$ kΩ: $C_b = \frac{228\times10^{-9}}{0.8473\times2200} \approx$ **122.3** pF
- For $R_p = 4.7$ kΩ: $C_b = \frac{485\times10^{-9}}{0.8473\times4700} \approx$ **121.8** pF

**Mean Estimated Bus Capacitance: $C_b \approx$ 121.5** pF. This represents the combined capacitance of the breadboard tracks, jumper wires, scope probe (15 pF), and MCU I/O pins.

# 4 Mandatory Task Questions & Detailed Solutions

**Task Question 4**

**At 400 kHz, find the largest pull-up resistor for which communication stays reliable. Does it agree with the theoretical calculations in Section 4.3?**

**Answer:** The theoretical upper limit of $R_p$ at 400 kHz ($t_{r(\max)} = 300$ ns) for our measured bus capacitance $C_b = 121.5$ pF is:

$$R_{p(\max)} = \frac{300\times10^{-9}\ \text{s}}{0.8473\times121.5\times10^{-12}\ \text{F}} = \textbf{2914}\blacksquare \approx \textbf{2.91}\ \text{k}\blacksquare \tag{2}$$

In practical bench testing, communication remained 100% error-free when using standard resistors up to **2.7** k■. When a 3.3 kΩ resistor was inserted, intermittent NACK errors began appearing on long data bursts. When a 4.7 kΩ resistor was used, the rise time reached 485 ns, completely corrupting the high-speed clock sampling. The experimental finding ($\leq$ 2.7 kΩ) agrees closely with the theoretical boundary of 2.91 kΩ.

**Task Question 5**

**Why is the rising edge slow and curved, but the falling edge is sharp and almost instantaneous?**

**Answer:** This fundamental asymmetry stems from the open-drain semiconductor architecture:

1. **Falling Edge (Active Pull-Down):** When a device transitions the line from HIGH to LOW, it turns ON its internal open-drain NMOS transistor. The NMOS transistor has an extremely small on-resistance ($R_{DS(on)} \approx$ 10–50 Ω). The bus capacitance $C_b$ discharges directly to ground through this low-impedance channel:

$$\tau_{\text{fall}} = R_{DS(on)} \times C_b \approx 30\,\Omega \times 120\ \text{pF} = \textbf{3.6}\ \text{ns} \tag{3}$$

   Because the discharge time constant is under 4 ns, the falling edge is virtually vertical.

2. **Rising Edge (Passive Pull-Up):** To transition the line from LOW to HIGH, the open-drain NMOS simply turns OFF (enters high-impedance mode). The line is not actively driven high; instead, current must trickle from the $V_{DD}$ rail through the external pull-up resistor $R_p$ to charge $C_b$:

$$\tau_{\text{rise}} = R_p \times C_b \approx 4700\,\Omega \times 120\ \text{pF} = \textbf{564}\ \text{ns} \tag{4}$$

   This exponential charging curve produces the characteristic slow, rounded RC rise profile.

# 5 Conclusion

Observing the physical signals demonstrates that the $I^2C$ bus is an analog circuit operating under digital rules. Ensuring that $R_p$ is properly sized relative to bus capacitance $C_b$ is critical to preventing edge degradation and ensuring high-speed communications reliability.